

Computational Astrophysics Exercises

Assignments Set 2 — Summer Term 2026

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Exercise 1: The vis-viva equation

1 pts

The specific energy ε in the two body problem is constant throughout the orbit. At any point:

$$\varepsilon = \frac{1}{2}\dot{r}^2 - \frac{\mu}{r} = \text{const} \quad (1)$$

At periapsis (r_p) and apoapsis (r_a), the velocity vector is perpendicular to (r). By conservation of specific angular momentum h :

$$h = r_p v_p = r_a v_a \rightarrow v_p = \frac{h}{r_p}, v_a = \frac{h}{r_a} \quad (2)$$

Equating energy at these two points:

$$\frac{1}{2} \left(\frac{h}{r_p} \right)^2 - \frac{\mu}{r_p} = \frac{1}{2} \left(\frac{h}{r_a} \right)^2 - \frac{\mu}{r_a} \quad (3)$$

$$\frac{h^2}{2} \left(\frac{1}{r_p^2} - \frac{1}{r_a^2} \right) = \mu \left(\frac{1}{r_p} - \frac{1}{r_a} \right) \quad (4)$$

$$\frac{h^2}{2} \left(\frac{r_a + r_p}{r_p r_a} \right) = \mu \quad (5)$$

Using the geometric relation for an ellipse $2a = r_a + r_p$:

$$\frac{h^2}{2} \left(\frac{2a}{r_p r_a} \right) = \mu \leadsto h^2 = \frac{\mu r_p r_a}{a} \quad (6)$$

Now we can substitute h^2 back into Equation 1 at periapsis to find the specific energy:

$$\varepsilon = \frac{h^2}{2r_p^2} - \frac{\mu}{r_p} = \frac{\mu r_p r_a}{2ar_p^2} - \frac{\mu}{r_p} = \frac{\mu r_a}{2ar_p} - \frac{\mu}{r_p} \quad (7)$$

$$\stackrel{r_a=2a-r_p}{=} \frac{\mu(2a-r_p)}{2ar_p} - \frac{\mu}{r_p} = -\frac{\mu}{2a} \quad (8)$$

Setting equal to Equation 1:

$$\frac{1}{2}\dot{r}^2 - \frac{\mu}{r} = -\frac{\mu}{2a} \quad (9)$$

Or:

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right) \quad (10)$$

Exercise 2: Kepler's equation and root finding

7 pts

(i)

We used the initial values from the hint: $E_0 = M$ for $e < 0.8$ and $E_0 = \pi$ for $e > 0.8$.

The orbits of Mercury ($e = 0.205$, $a = 0.39$ au) and Halley's comet ($e = 0.967$, $a = 17.8$ au) are plotted below.

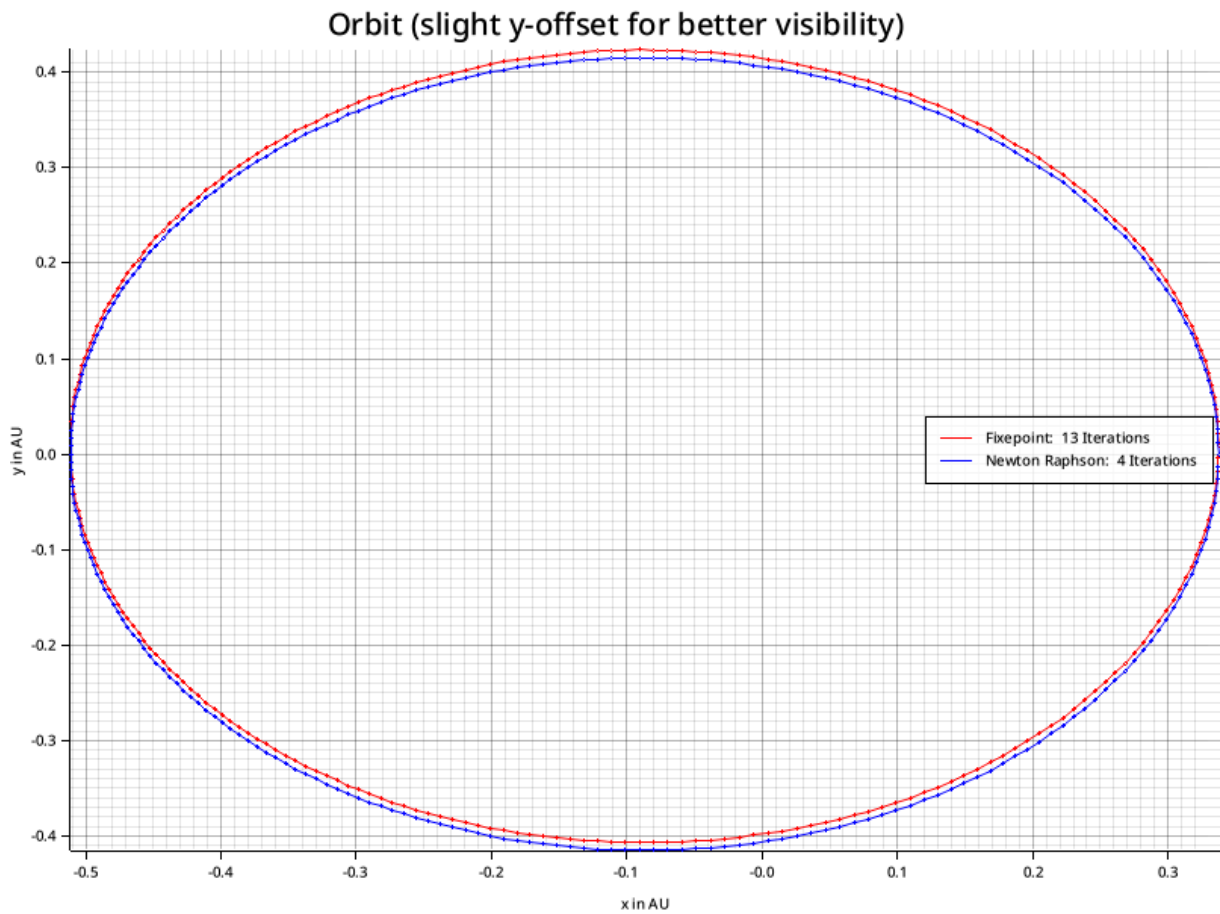


Figure 1: Orbit of Mercury. Both methods agree, but Newton–Raphson converges in fewer iterations.

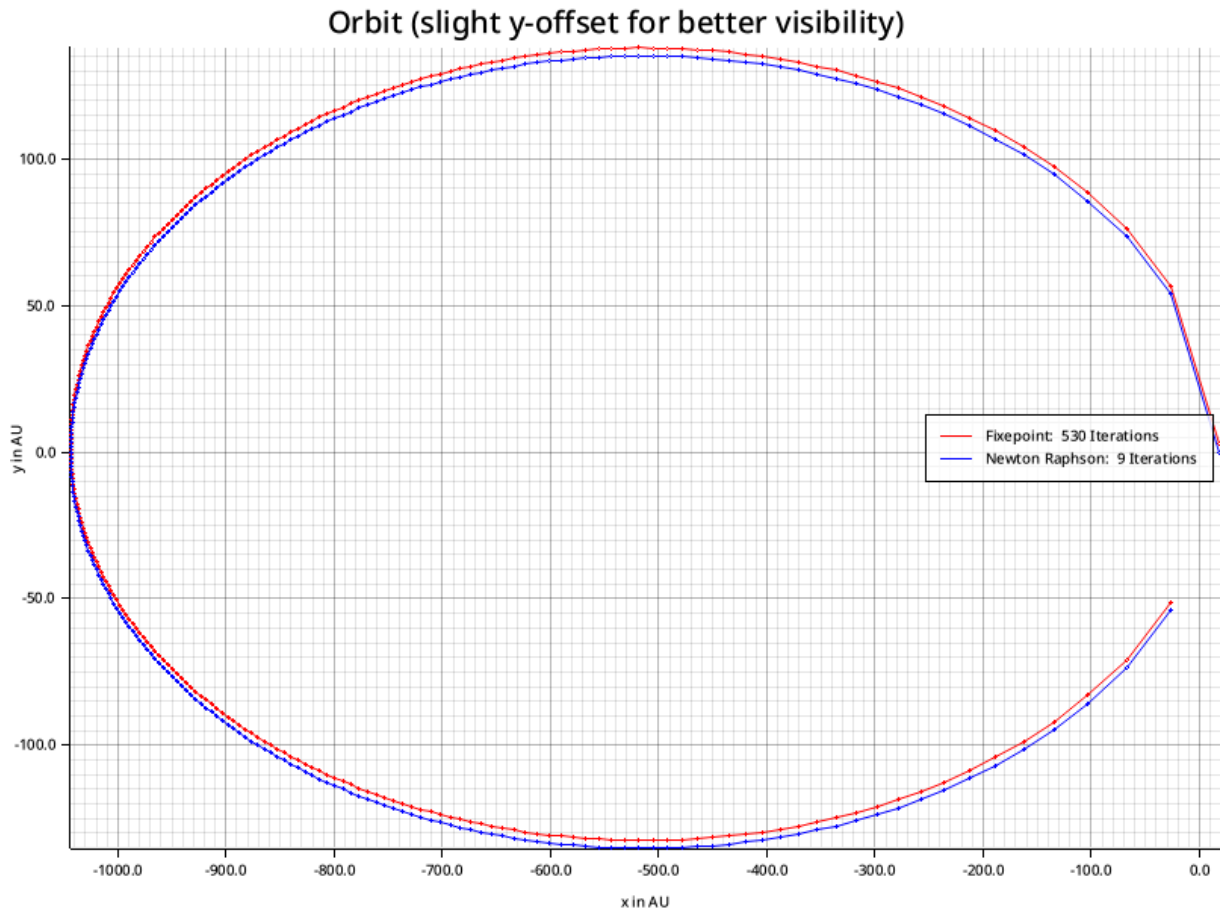


Figure 2: Orbit of Halley's comet. The high eccentricity makes the fixed-point iteration converge more slowly; Newton–Raphson remains efficient.

Newton–Raphson converges quadratically and therefore requires significantly fewer iterations than the linearly-convergent fixed-point scheme, especially for the more eccentric Halley orbit.

(ii) Using the J2000 orbital elements, we determine the mean anomaly of Earth and Mars on 1 January 1985 via

$$M_0(\text{planet}) = \lambda - \varphi_0 - 15 \text{ yr} \cdot \frac{2\pi}{P}, \quad (11)$$

then propagate forward day by day with

$$M(t) = \frac{2\pi t}{P} + M_0, \quad (12)$$

solving Kepler's equation at each step with Newton–Raphson. The Euclidean distance between the two positions is plotted below.

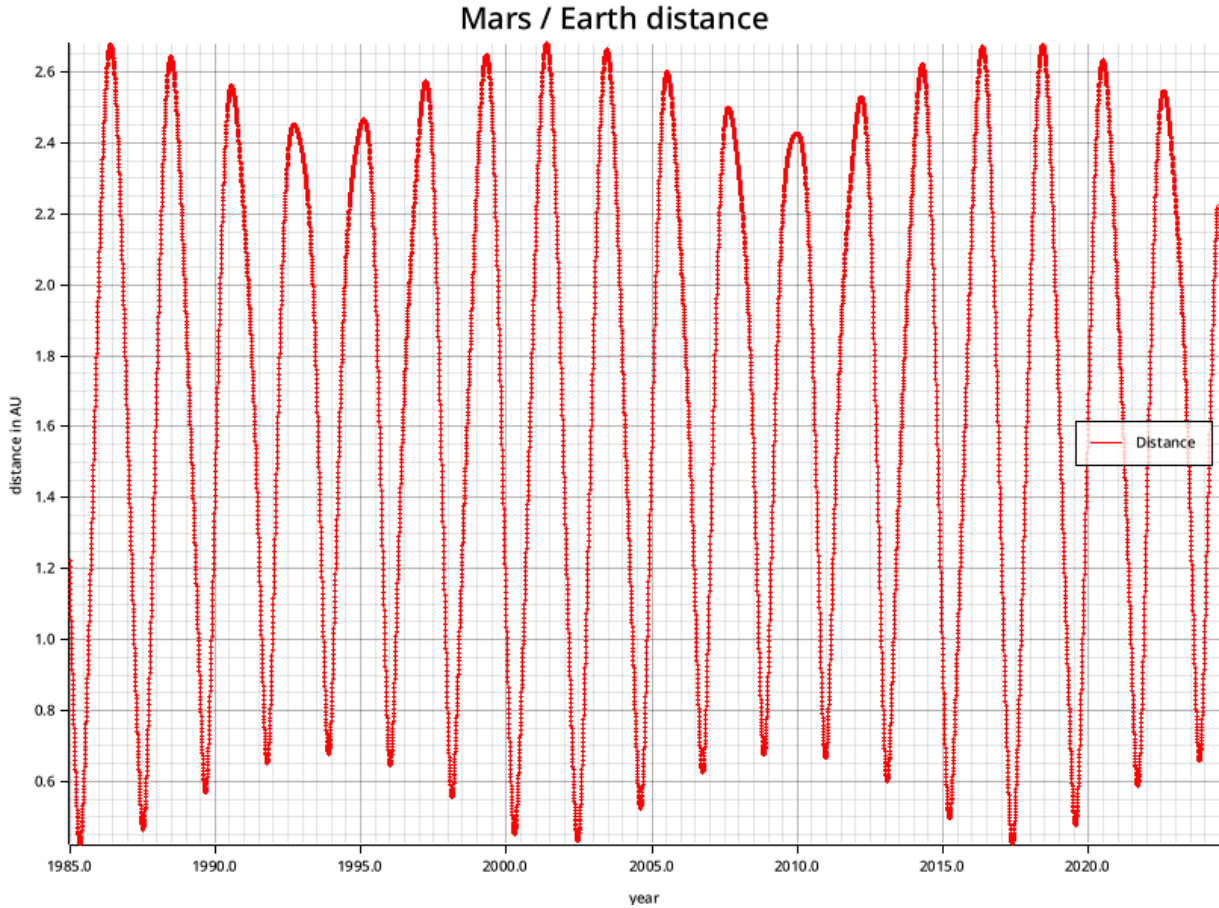


Figure 3: Earth–Mars distance from 1 January 1985 to 29 May 2024. The roughly 26-month period where the planets are closest is visible as repeating minima

Exercise 3: The Lagrange point

2 pts

The L1 Lagrange point satisfies

$$\frac{GM}{r^2} - \frac{Gm}{(R-r)^2} = \omega^2 r \quad (13)$$

or equivalently $f(r) = 0$ with

$$f(r) = \frac{GM}{r^2} - \frac{Gm}{(R-r)^2} - \omega^2 r. \quad (14)$$

We have implemented Newton–Raphson directly in Rust with

$$f'(r) = -\frac{2GM}{r^3} - \frac{2Gm}{(R-r)^3} - \omega^2. \quad (15)$$

Using the given parameters with starting value $r_0 = \frac{R}{2}$, Newton–Raphson converges to a distance from the earth of

$$r_{L1} \approx 3.260 \times 10^8 \text{ m} \quad (16)$$

or roughly 5.84×10^7 m in front of the Moon.